

How the Answer Is Built

From question to response — the complete picture

We've covered how your question travels to the model and how attention works. Now the question everyone actually has: **Where does the answer come from — and how does it get written?**

The Question

"What is my portfolio's exposure to emerging-market equities?"

The Answer

"Your portfolio has 12.4% exposure to emerging-market equities, primarily through the Vanguard FTSE EM ETF."

The answer has two completely separate parts. Understanding them separately is the key to understanding LLMs.

The Substance

The number 12.4%, the fund name. **What** is being said.

The Framing

"Your portfolio has X% exposure to..." **How** it is being said.

They come from different places. Let's take them one at a time.

Part 1 — Where the Substance Comes From

The substance of any answer — the actual facts, figures, and specific details — can only come from one of three places:

1

Source A: Something in the Context Window

The tool result, the RAG document, something the user stated earlier in the conversation. It's sitting right there in the context window as text.

In our example, the tool returned: {
"exposure_pct": 12.4, "primary_holding":
"Vanguard FTSE EM ETF" }

That text is in the context. Attention connected it to the question. When the model writes "12.4%", it is pattern-completing from something that genuinely exists in front of it. This is a **grounded answer** — traceable to a real source.

2

Source B: Something Learned During Training

The model read billions of documents before it ever spoke to anyone. General knowledge, definitions, how markets work, how sentences are structured in finance — all of this was absorbed during training and lives in the model's weights.

If you asked "*What does emerging-market mean?*" — there's no tool result for that. The model answers from training. This is also legitimate, as long as the knowledge hasn't changed since the training cutoff.

3

Source C: Neither — Hallucination

No tool. No RAG. Nothing in the context. No reliable training pattern. The model still generates *something* — because that's all it knows how to do. It pattern-completes with whatever digit sequence or phrase sounds most plausible given the context.

The result looks exactly like Sources A and B. Same confident tone. Same professional framing. But the substance was fabricated.

 **This is the single most important thing to understand about LLMs.** The model has no internal alarm that says "I don't actually have this information." The generation loop runs identically whether the answer is grounded or made up. You cannot tell from the output which source it came from — only from knowing whether a tool was called or a relevant document was retrieved.

Part 2 — Where the Framing Comes From

Now separately: where does "Your portfolio has X% exposure to emerging-market equities, primarily through..." come from?

Not from the Tool

The tool returned a JSON object with a number — nothing more.

Not from a Document

The RAG chunk might have been a research note about EM markets.

From Training

The sentence structure, the professional tone, the way a wealth-management answer is shaped — comes entirely from training.

During training, the model read millions of examples of how financial professionals write. How advisors answer portfolio questions. How sentences like this are structured. It absorbed the *pattern* of a good answer in this domain.

Think of it like a new analyst joining your firm. On day one, they don't know your clients' numbers. But they already know how to write a professional client note — structure, tone, appropriate caveats — from everything they studied and observed before. The model is the same: it brings the framing from training, and fills in the substance from the context.

Substance and framing are independent. The model can have:

✓ Right substance, right framing

Tool was called, model is well-trained.

Right substance, wrong framing

Tool was called, but tone is off.

Wrong substance, right framing

Hallucinated number, professional sentence.

Wrong substance, wrong framing

Hallucinated everything.

⊗ The framing looking professional tells you **nothing** about whether the substance is correct.

Part 3 — Attention's Role in Both

Here is where attention fits — and it fits in **both** places.

Attention While Reading the Question

When the question arrives, attention runs across the entire context window — question, system prompt, tool results, RAG chunks. Every token looks at every other token simultaneously.

- *"Exposure"* figures out it means financial exposure, not sunlight or photography.
- *"My"* attaches to the portfolio.
- *"Emerging-market equities"* locks together as one concept.
- *"12.4"* in the tool result gets connected to the question being asked.

By the end of this, the model has a deep understanding of what is being asked and what relevant information exists in the context. This is the **input pass** of attention.

Attention While Writing the Answer

Attention doesn't stop once writing begins. Every time the model predicts a new word, attention runs again — this time across:

- **Everything in the input** (the question, tool results, system prompt)
- **Everything already written in the answer so far**

So when the model is about to write the word after *"Your portfolio has 12.4%"*, it attends to both the growing answer AND the original question. The word *"exposure"* wins partly because the question used that word, and partly because *"has 12.4% exposure"* is the natural pattern in professional finance writing.

 **The question stays alive throughout the entire answer.** It doesn't just trigger the generation and disappear. Every word the model writes is still being shaped by attention reaching back to the question.

Part 4 — Now the Generation Loop

With that foundation, the generation loop makes complete sense. The model has:

01

02

03

Read and Understood the Question

Input attention pass completed.

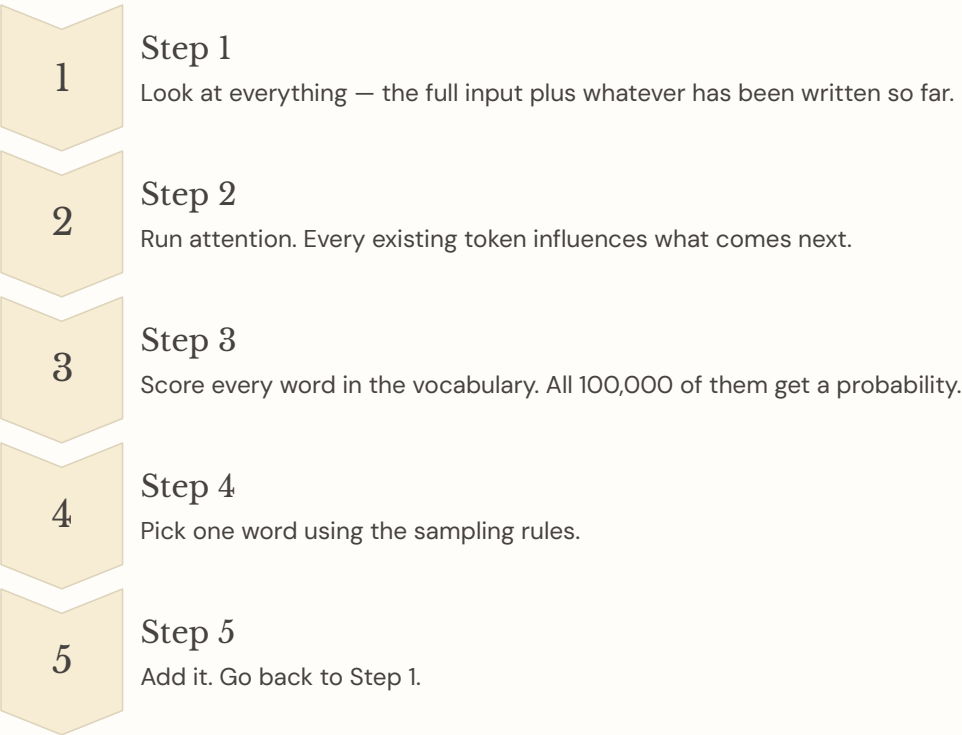
Connected the Tool Result

What's being asked is linked to what's available.

Drawn on Training

How this kind of answer should be shaped is ready.

Now it writes. One token at a time. The same five-step loop, repeated until done:



Let's watch it happen — word by word.

The Answer Being Written — Word by Word

The percentages below are illustrative. The actual numbers live inside a GPU as floating-point values no human ever reads. The mechanism is real; the specific numbers show the shape of the reasoning.

- We will trace the construction of: **"Your portfolio has 12.4% exposure to emerging-market equities, primarily through the Vanguard FTSE EM ETF."** — one token at a time, watching attention and probability at every step.



Word 1 — "Your"

The model attends to the full context. Nothing written yet. Training patterns for wealth-management answers surge to the top.

Candidate	Probability	Why
"Your"	44%	Wealth-management answers to "what is my X" almost always begin with "Your X..."
"Based"	14%	"Based on your holdings..." — valid but indirect
"The"	11%	Impersonal — less appropriate for a direct client answer
"I"	9%	"I can see..." — possible but less direct
Others	22%	—

Picks: "Your"

Answer so far: **Your**

Word 2 — "portfolio"

"Your" is now in the context. It's a possessive — something belongs to the person. The question was about a portfolio. Attention pulls "portfolio" forward strongly.

Candidate	Probability	Why
"portfolio"	71%	The question was about a portfolio — "Your portfolio" is the natural pair
"exposure"	9%	Possible but off-order
"holdings"	6%	Valid alternative
Others	14%	—

Picks: "portfolio"

Answer so far: **Your portfolio**

Word 3 — "has"

"Your portfolio" is a subject. A verb follows. The factual answer pattern — "Your portfolio *has* X%" — dominates.

Candidate	Probability	Why
"has"	58%	"Your portfolio has X% exposure" — the most natural factual structure
"currently"	14%	Adverb — adds a word without adding meaning
"shows"	11%	Slightly informal
"holds"	9%	Changes sentence structure
Others	8%	—

Picks: "has"

Answer so far: **Your portfolio has**

Word 4 — The Most Important Step

Context: **Your portfolio has**

This is where substance meets framing. The tool result is sitting in the context window as plain text:

```
{ "exposure_pct": 12.4, "primary_holding": "Vanguard FTSE EM ETF" }
```

Attention runs across everything — including those characters. The model has seen countless training examples of sentences like “Your portfolio has [number]% exposure” following a tool result containing that number. So attention pulls “12.4” from the tool result forward, and the probability of outputting “12” next becomes very high.

 The model is **not copying the number**. It is not doing arithmetic. It is pattern-completing — the tool result is in the context, attention connected it to the gap in the sentence, and the generation loop fills the gap with what training says belongs there.

Candidate	Probability	Why
" 12"	67%	"12.4" is in the tool result. Attention linked it here.
" approximately"	11%	System prompt says cite directly — hedging loses
" a"	8%	"has a significant..." — vague
Others	14%	—

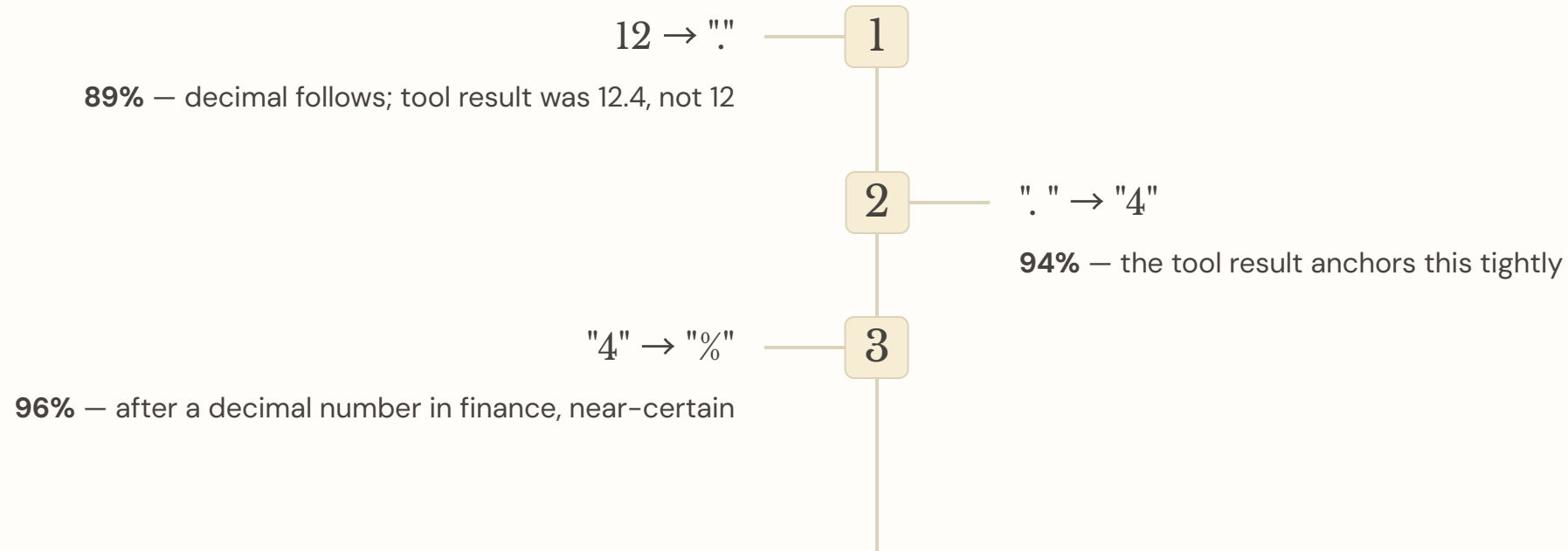
Picks: " 12"

Answer so far: **Your portfolio has 12**

 **The number arrives in pieces.** “12.4%” will take four separate prediction steps: 12 → . → 4 → %. Each piece is its own token, its own probability contest. The model predicts each digit in sequence — not as a number, but as the next most likely character. This is why LLMs make arithmetic mistakes: they predict digits by pattern, not by calculation. There is no calculator inside.

Words 5, 6, 7 — The Number Completes

The tool result anchors each digit tightly. Watch the probabilities climb as the pattern locks in:



Answer so far:

Your portfolio has 12.4%

Three tokens. Three separate probability contests. Each one won by the pattern anchored in the tool result sitting in context. The number is complete — but the sentence is not.

Word 8 — "exposure"

Number complete. Attention pulls back to the question — which used the word "exposure." That word now wins clearly.

Candidate	Probability	Why
" exposure"	61%	The question used "exposure" — attention reaches back and mirrors it
" allocated"	16%	Valid but changes the sentence shape
" invested"	10%	Similar
" in"	8%	Shorter but less formal

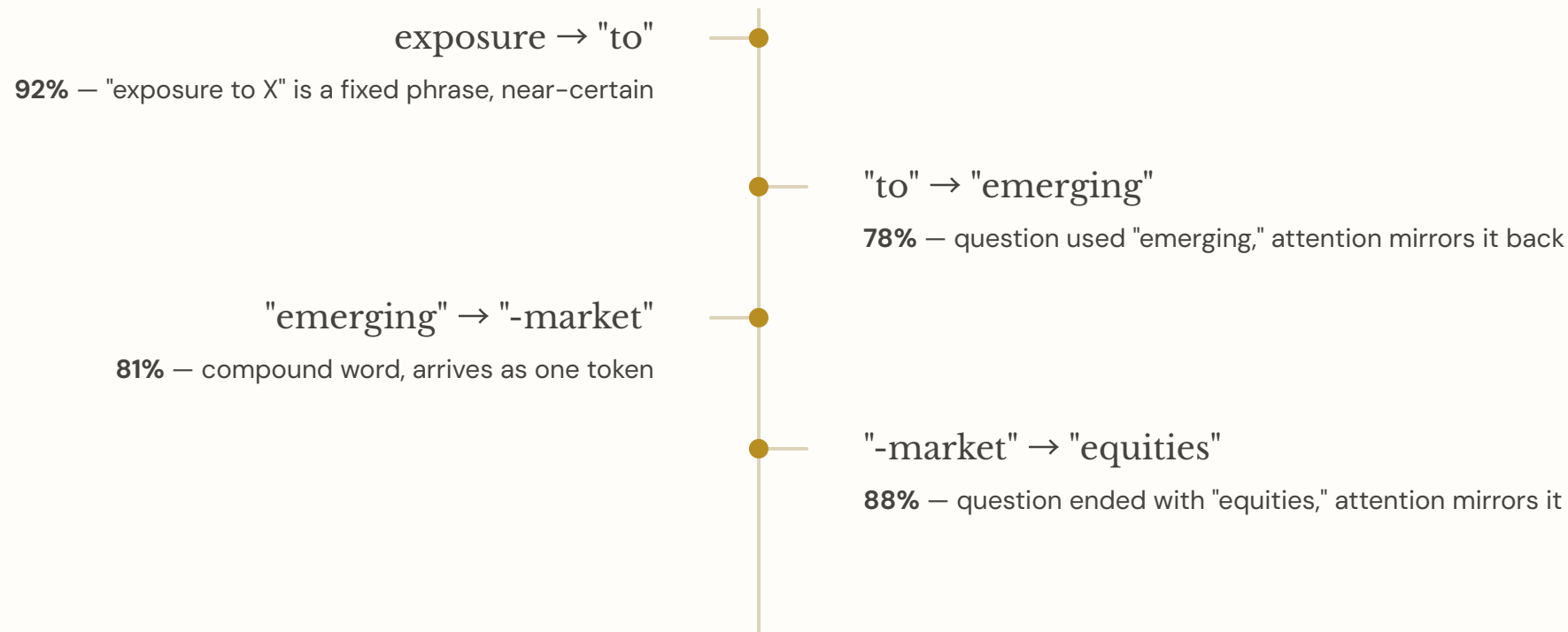
Picks: " exposure"

Answer so far: **Your portfolio has 12.4% exposure**

- ✔ **Notice what just happened.** The word "exposure" in the answer came from the word "exposure" in the *question* — via attention reaching back across the entire input during output generation. This is attention doing its second job: keeping the answer tethered to the question while writing.

Words 9 Through 12 — The Destination

The phrase locks in, token by token, each one mirroring the question's own language back through attention:



Answer so far:

Your portfolio has 12.4% exposure to emerging-market equities

Every word in this phrase was seeded by the question itself. Attention during output generation kept reaching back, pulling the question's vocabulary into the answer's vocabulary — word for word.

Word 13 — End or Elaborate?

The core answer is complete. The system prompt said "lead with the number, then explain." The number has been led with. The model leans toward one elaborating clause before stopping.

Candidate	Probability	Why
","	44%	Continue with elaboration
."	38%	End cleanly
" primarily"	11%	Begin elaboration directly
" —"	5%	Em-dash elaboration

Picks: ","

The model adds: *"primarily through the Vanguard FTSE EM ETF."* — pulled from the tool result still sitting in context. Then it predicts "." and then the end token. Generation stops.

A photograph of a wooden desk with a document and a pen. The document is a multi-page paper with dense, illegible text. A black pen lies diagonally across the bottom right of the document. The scene is lit with warm, low-key lighting, creating a professional and focused atmosphere.

The Completed Answer

"Your portfolio has 12.4% exposure to emerging-market equities, primarily through the Vanguard FTSE EM ETF."

Substance

12.4% and Vanguard FTSE EM ETF — came from the **tool result**, via attention.

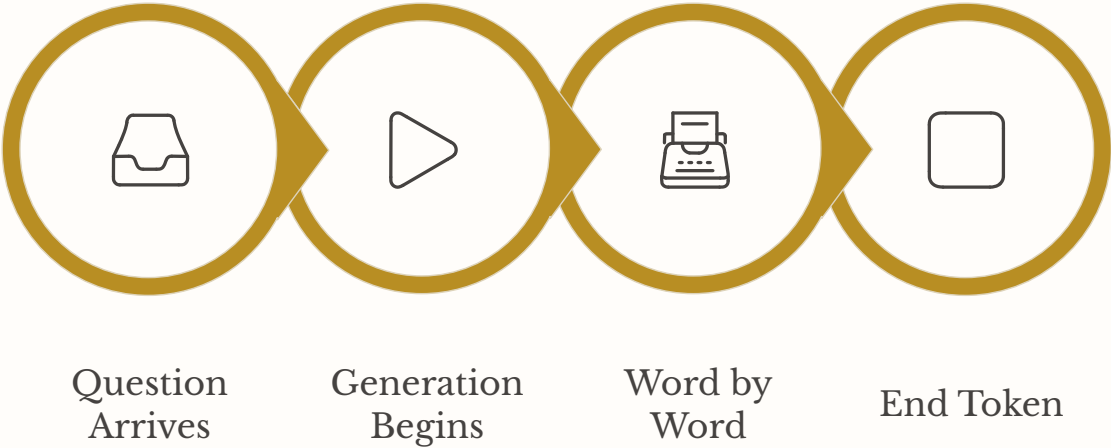
Framing

"Your portfolio has X% exposure to Y, primarily through Z" — came from **training patterns**.

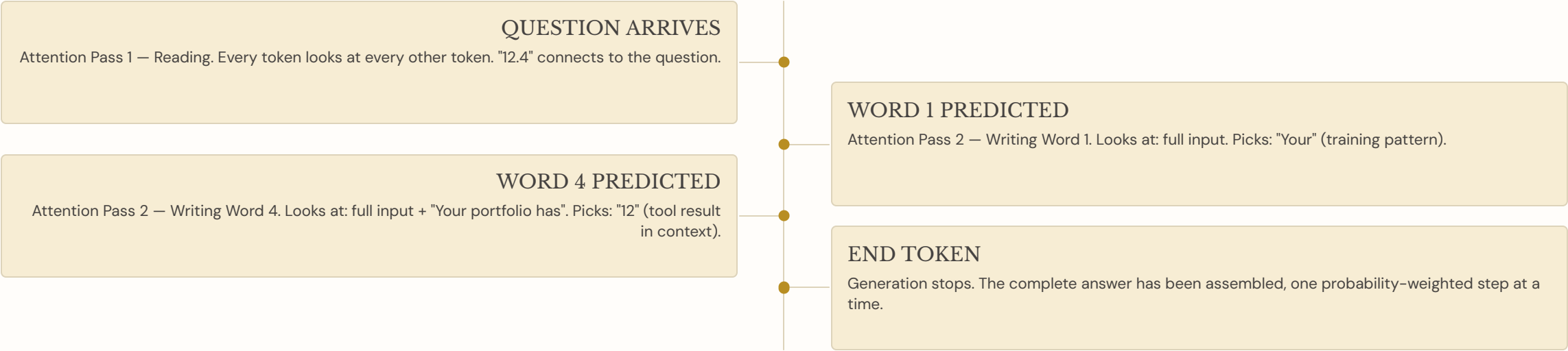
Every Word

Shaped by attention running across both the input and the growing answer at **every single step**.

The Picture in Full



At every stage, attention is doing work — first to understand, then to write. The question never disappears from the model's view. The tool result stays in context throughout. Training patterns shape every sentence. And the loop runs, one token at a time, until the end token wins.



Three Things to Carry Into Any Meeting About AI

1

The framing looking good tells you nothing about the substance being correct

A hallucinated answer and a grounded answer look identical. Same professional tone. Same sentence structure. Same confidence. The only difference is whether the substance traces back to something real in the context. This is why tools and RAG aren't optional — they're the only mechanism that puts real substance into the context for attention to work with.

2

Your words shape the answer's words — directly

"Equities" won at Word 12 because you used "equities" in your question.

"Emerging" won at Word 10 because you used "emerging." Attention during output generation keeps reaching back to your question. This is why how you phrase a question matters — you're not just communicating intent, you are literally seeding the probability tables for every word the model will write in return.

3

There is no answer before the first word is written

There is no draft. No plan. No pre-composed response waiting to be delivered. The model starts from zero and commits to one word at a time — each word changing what the next word will be. The coherent answer that results is an emergent property of the pattern, not a pre-generation decision. This is also why you can't ask an LLM to "think first and then answer" without reasoning models — for a standard model, thinking *is* the writing.

The One-Paragraph Version

Before the model writes a single word, attention has already done two things: connected the question to what matters in the context, and pulled the relevant substance — the 12.4% from the tool result — into focus. Then generation begins. One token at a time. For each token, attention runs again — this time across the full input plus everything already written — and the most probable next word wins. "Your" comes from training patterns. "12.4%" comes from the tool result via attention. "Exposure" comes from the question itself, mirrored back. The framing comes from training; the substance comes from the context. They combine, word by word, into a sentence that looks planned but was assembled one probability-weighted step at a time, from nothing, to the full stop.

Substance

From the context window — tool results, RAG, conversation history.



Framing

From training — billions of documents, patterns, professional language.



Attention

The mechanism that connects them — reading and writing, every step.